2. Overview

This week's lab includes two console-based Python programs: a Password Strength Checker and an Email Validation System. Both use built-in string methods for validation and feature a common boxed-menu interface.

PROGRAM 1 — PASSWORD STRENGTH CHECKER

2.1 Introduction and Background

Weak passwords are a major security risk. This program develops a console-based Password Strength Checker in Python that evaluates passwords using predefined rules and provides clear feedback. It focuses on rule-based string validation and character analysis using Python's built-in string methods.

2.2 Objectives

- Develop a console-based Password Strength Checker.
- Validate password length, uppercase, lowercase, digits, and special characters.
- Classify passwords as Weak, Medium, or Strong.
- Display a strength bar and requirement checklist.
- Practice Python string validation methods.
- Test the program with different password combinations.

2.3 Problem Statement / Driving Question

How can a password be evaluated using simple rule-based checks to provide clear feedback and classify its strength?

2.4 Methodology

Program Structure: Used reusable box-drawing helper functions to create a consistent bordered menu and result interface with an ASCII-art banner.

Password Validation Logic: Developed password_validator() to check password length, digit, lowercase, uppercase, and special character requirements, then classified passwords as weak, medium, or strong.

Result Presentation: Displayed a strength bar, an **[OK]/[NO]** requirement checklist, and formatted feedback inside the bordered interface.

Menu-Driven Control Flow: Implemented a continuous menu with **Check Password** and **Exit** options, and tested the program with different password combinations to verify correct classification.

2.5 Expected Outcomes / Deliverables

- A functional Password Strength Checker.
- Password classification with visual feedback.
- A user-friendly boxed console interface.
- Practical understanding of rule-based string validation.
- Scope for future improvements such as entropy-based scoring and common-password detection.